Model and Experiment

Building Models to Describe our World

Theories, Uncertainty, and Dimensions

Outline

Building Models
- Model vs experiment
- Does my prediction make sense?
- Base dimenions and units

Uncertainty and Error
- Uncertainties
- Precision vs accuracy vs errors
- Propogating uncertainties

Dimensional Analysis

Building Models

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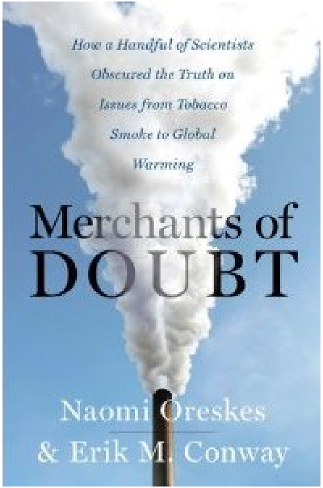
Theories, Uncertainty, and Dimensions

Model and experiment

- We wish to test a theory, so we plan an experiment. That's the fundamental aspect of science!
- We use the theory to build a model and predict the result that we expect from our experiment
- We compare the experimental result to the model prediction. Neither can ever be known with infinite precision:
    - Model prediction depends on the assumptions that were made and uncertainties in the parameters of the model
    - Measurement depends on experimental uncertainties

→ Science can never make definitive statements, only statements based on probabilities (the probability that global warming is not due to human activity is very small, the probability that smoking does not cause lung cancer is very small. . .).



Does my model prediction make sense?

- Did I correctly apply the rules for using a given theory? (e.g. identify all of the forces on an object, when applying Newton's theory)
- Does my predicted value have the correct order of magnitude? (e.g. did I predict that a ball will fall faster than the speed of light?)
- Does my predicted value have the correct dimension? (e.g. did I predict that it would take a time of 3m to fall a distance of 1m?)
- Does my predicted value change as I expect that it would? (e.g. if I drop a ball from higher, does my predicted fall time increase?)

→ **In assignments, we will require you to include a few sentences to evaluate whether your predictions make sense.**

Base dimensions and units

- Anything that you can measure/quantify has a dimension.
- Multiple units can be used for the same dimension (e.g mm, ft, soccer fields are all units for the dimension of length, L).
- We can define a set of Base Dimension, and any other dimension can always be expressed in terms of those base dimensions. (e.g the dimension of speed is L/T)
- Base SI units are the SI units that correspond to the base dimensions. (e.g. the SI unit for speed is m/s)

Dimension	**SI unit**
Length [L]	meter [m]
Time [T]	seconds[s]
Mass [M]	kilogram [kg]
Temperature [Θ]	kelvin [K]
Electric current [I]	ampère [A]
Amount of substance [N]	mole [mol]
Luminous intensity [J]	candela [cd]
Dimensionless [1]	unitless []

Base dimensions and their SI units with abbreviations.

You don't need to memorize these, you will know them with practice (in general, don't memorize things in physics, just ask if we forgot something on the formula sheet!)

Algebra with dimensions/units

- If you obtain a formula for your model prediction, that prediction will have dimensions, based on the inputs to your model
- For example, distance, x, in terms of an acceleration, g, initial speed, v_0, and time, t:

$$x = v_0 t + \frac{1}{2} g t^2 \tag{1}$$

- Evaluating the dimension of x:

$$[x] = [v_0][t] + \left[\frac{1}{2}\right][g][t^2] = \frac{L}{T}T + 1\frac{L}{T^2}T^2 = L + L \tag{2}$$

- You can use the SI units instead of the dimensions, to carry out the same exercise
- **Always check that your prediction is dimensionally correct!** It won't guarantee that your formula is correct, but it will identify if it is definitely wrong!

Uncertainty and Error

Measurement uncertainty

- You can never measure anything to infinite precision.
- You can only ever say that a value is likely within some range.
- We communicate this as:

$$x = (10.0 \pm 0.1)\text{m} \quad \rightarrow \quad \text{“central value”} \pm \text{“uncertainty”} \tag{3}$$

- We mean that we are *pretty confident* that x is between 9.9 m and 10.1 m.
- It is up to us, scientists, to **be precise** in explaining what we mean by *pretty confident* $\rightarrow$ usually, we are 68% confident. Ultimately, we need to be precise in describing how we determined the central value and uncertainty.

Relative uncertainty

- We want to know how precise our measurement is.
- In other words, is the uncertainty large? Are we confident that the true value is close to the central value?
- We quantify how large the uncertainty is relative to the central value $\rightarrow$ relative uncertainty:

$$x = \bar{x} \pm \sigma_x$$

$$\text{Relative uncertainty} = \frac{\sigma_x}{\bar{x}} = \frac{\text{uncertainty}}{\text{central value}}$$

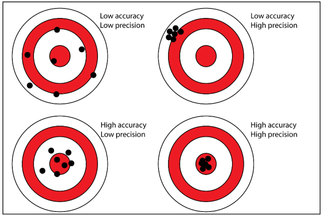
- Express the relative uncertainty in percent
- Usually, we consider 1% relative uncertainty as precise
- You measured that the Moon is 1 ± 1 light years from the Earth; ok, good for you, but your very imprecise measurement is not very useful to test any model!

Determining an uncertainty

- This is by far the trickiest part of experimental physics, and usually the main focus of any experiment.
- There is no "set in stone" rule for determining uncertainties, but there are some guidelines and accepted methods.
- **The most important part is that you precisely describe (and justify) how you determined an uncertainty, and that you make every effort to minimize it.**
- Ideally, if you can repeat a measurement multiple times; then this is the best way to determine an uncertainty:
    - Mean and standard deviation (you'll get practice in the labs next week)
    - Mean is the average, standard deviation is the average distance of measurements to the mean (whether or not the measurements are very scattered about the mean)
- If appropriate, you can use half the smallest division of your measuring device as the uncertainty (it should rarely be smaller than this).

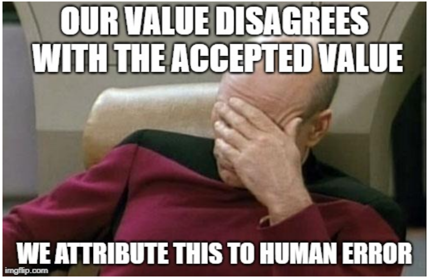
Precision and accuracy, random and systematic errors

- You might find that multiple measurements have a small spread (and result in a small uncertainty) $\rightarrow$ Your measurements are *precise*.
- You later discover that your measurements are "way off" from the true value $\rightarrow$ Your measurements are not *accurate*.
- **Random error:** the spread in values that you get when repeating a measurement.
- **Systematic error:** the offset of your measurement from the "true" value. In "real" science, this is very difficult to detect, since we never know the true value.
- In first year physics, if you measure $g = 5 \pm 1\mathrm{m/s}^2$, there is likely a systematic error



Human error

- **That's not a thing. Don't use that term!**
- Do you mean that you made a mistake? Then fix it!
- Do you mean that, due to reaction time, you have a systematic uncertainty in your time measurement? Then include that error!
- Do you mean that, due to parallax, you have a systematic uncertainty in your length measurement? Then include that error!



TAs grading lab reports

Propagating uncertainties

- You measured, say, distance, x, and time, t, and want to know speed, v:

$$x = \bar{x} \pm \sigma_x$$
$$t = \bar{t} \pm \sigma_t$$
$$v = v(x,t) = \frac{x}{t} = \bar{v} \pm \sigma_v$$

- You want to propagate the uncertainties in x and t to determine the uncertainty in v
- Derivative method + adding in quadrature:

$$\bar{F} = F(\bar{x}, \bar{y})$$
$$\sigma_F = \sqrt{\left(\frac{\partial F}{\partial x}\sigma_x\right)^2 + \left(\frac{\partial F}{\partial y}\sigma_y\right)^2}$$

When $F(x,y) \to v(x,t)$

- Central value:

$$\bar{v} = \frac{\bar{x}}{\bar{t}}$$

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$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}\frac{x}{t} = \frac{1}{t}$$

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\begin{aligned}
\sigma_v &= \sqrt{\left(\frac{\partial v}{\partial x}\sigma_x\right)^2 + \left(\frac{\partial v}{\partial t}\sigma_t\right)^2} \\
\frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}\frac{x}{t} = \frac{1}{t} \\
\frac{\partial v}{\partial t} &= \frac{\partial}{\partial t}\frac{x}{t} = \frac{-x}{t^2} \\
\therefore \sigma_v &= \sqrt{\frac{1}{\bar{t}^2}\sigma_x^2 + \frac{\bar{x}^2}{\bar{t}^4}\sigma_t^2}
\end{aligned}
$$

- phew!

Simple prescriptions

- The derivative rule leads to simple prescriptions in most cases.
- In practice, we'll show you how to use a computer and QExpy to propagate uncertainties (see Appendix D).

Operation to get z	**Uncertainty in z**
$z = x + y$ (addition)	$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$
$z = x - y$ (subtraction)	$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$
$z = xy$ (multiplication)	$\sigma_z = xy\sqrt{\left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2}$
$z = \frac{x}{y}$ (division)	$\sigma_z = \frac{x}{y}\sqrt{\left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2}$
$z = f(x)$ (a function of 1 variable)	$\sigma_z = \left\lvert\frac{df}{dx}\sigma_x\right\rvert$

How to propagate uncertainties from measured values $x \pm \sigma_x$ and $y \pm \sigma_y$ to a quantity $z(x, y)$ for common operations. Addition and subtraction: use absolute uncertainties. Multiplication and division, use relative uncertainties.

Dimensional Analysis

Building Models to Describe our World

Theories, Uncertainty, and Dimensions

Dimensional analysis

- Sometimes, you know that a certain quantity, z, must depend on other quantities, x and y, but you have no idea how to model it:

$$z = f(x, y) = x^a y^b$$

- As a minimum, you know that you must combine x and y in a way that the combination has the dimensions of z, but you want to know the powers, a and b.
- Suppose that:

$$[z] = \frac{L}{T^2}, \quad [x] = M \quad [y] = \frac{ML}{T^2}$$

Determining a and b

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$\rightarrow$ $b = 1$ and $a = -1$. $f(x, y) = x^a y^b$ must thus be of the form:

$$z = x^{-1} y^1 = \frac{y}{x}$$